

Time Series

Spring Semester 2026, EPFL
École Polytechnique Fédérale de Lausanne
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Practice Exam — Week 5: Trends, Seasonality & SARIMA

Duration: 60 minutes · No calculator · Answer on paper

Name: _____ Surname: _____

SCIPER (Student Card Number): _____

Exercise	Points
1	
2	
3	
4	
Total	

Justify every answer. Throughout, ε_t is a mean-zero white noise of variance σ^2 .

Problem 1 (5 points)

Consider the signal-plus-noise model $X_t = \mu_t + Y_t$, where μ_t is a deterministic trend and $\{Y_t\}$ is a zero-mean stationary process with autocovariance sequence γ_τ . Recall the difference operator $\nabla = I - B$ and the seasonal difference $\nabla_{(s)} = I - B^s$.

(a) Take $\mu_t = a + bt$ (linear trend). Show that $\nabla X_t = b + \nabla Y_t$, and explain in one sentence why ∇Y_t is stationary while $\{X_t\}$ in general is not. (1.5 pts)

(b) Now let $\mu_t = a + bt + s_t$ with a *quarterly* season $s_{t+4} = s_t$. Compute $\nabla \nabla_{(4)} X_t$ *explicitly* as a linear combination of the Y 's, and verify that its mean is 0 and that it is weakly stationary. (2 pts)

(c) (*A classic trap.*) The operators $\nabla_{(s)} = I - B^s$ and $\nabla^s = (I - B)^s$ are *not* the same. For the deterministic sequence $\mu_t = t^2$, compute both $\nabla^2 \mu_t$ and $\nabla_{(2)} \mu_t$ and state which kind of structure each operator removes. (1.5 pts)

Problem 2 (5 points)

Let $\{X_t\}$ be the multiplicative seasonal process $\text{SARMA}(0, 1) \times (1, 0)_4$ (quarterly data), defined by

$$(1 - \Phi B^4)X_t = (1 - \theta B)\varepsilon_t, \quad |\Phi| < 1,$$

a seasonal autoregression at lag 4 driven by a non-seasonal $\text{MA}(1)$ input.

(a) Write the explicit one-step recursion for X_t , and by inverting the seasonal AR factor obtain the $\text{MA}(\infty)$ weights ψ_j in $X_t = \sum_{j \geq 0} \psi_j \varepsilon_{t-j}$ (give a closed form for ψ_{4l} and ψ_{4l+1} , and state that all other $\psi_j = 0$). (2 pts)

(b) Using the white-noise filter rule $\gamma_\tau = \sigma^2 \sum_{j \geq 0} \psi_j \psi_{j+\tau}$, compute $\gamma_0, \gamma_1, \gamma_3, \gamma_4, \gamma_5$ in closed form, and explain why the autocovariance lives in clusters around the seasonal lags $0, 4, 8, \dots$ (2 pts)

(c) Evaluate the autocorrelations ρ_1, ρ_3, ρ_4 numerically for $\Phi = \frac{1}{2}$, $\theta = \frac{2}{5}$, $\sigma^2 = 1$. (1 pt)

Problem 3 (5 points)

This problem studies non-stationary integrated models.

(a) (*Random walk.*) Let $X_t = \sum_{j=1}^t \varepsilon_j$ with $X_0 = 0$ (an ARIMA(0, 1, 0)). Compute $\mathbb{E}[X_t]$, $\text{Var}(X_t)$ and $\text{Cov}(X_t, X_{t+\tau})$ for $\tau \geq 0$, deduce that $\{X_t\}$ is *not* stationary, and show ∇X_t is white noise. Show also that $\text{Corr}(X_t, X_{t+\tau}) \rightarrow 1$ as $t \rightarrow \infty$ with τ fixed. (2 pts)

(b) (*IMA(1, 1).*) Let $Y_t = Y_{t-1} + \varepsilon_t - \theta\varepsilon_{t-1}$ with $Y_0 = 0$. By unrolling the recursion, show that for $t \geq 1$

$$Y_t = \varepsilon_t + (1 - \theta) \sum_{j=1}^{t-1} \varepsilon_j - \theta\varepsilon_0,$$

and hence compute $\text{Var}(Y_t)$. For $\theta \neq 1$, what is the order of growth of $\text{Var}(Y_t)$ in t ? (2 pts)

(c) (*Over-differencing.*) What happens to Y_t and to $\text{Var}(Y_t)$ at the boundary $\theta = 1$? Show that ∇Y_t is then a *non-invertible* MA(1), and state the value of its lag-1 autocorrelation ρ_1 . Why is this the warning signature of over-differencing? (1 pt)

Problem 4 (5 points)

(Revision of Week 4: Yule–Walker estimation.) Let $\{X_t\}$ be a mean-zero causal AR(2) process $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$ observed at $X_1, \dots, X_n$, with sample autocovariances

$$\hat{\gamma}_0 = 10, \quad \hat{\gamma}_1 = 5, \quad \hat{\gamma}_2 = 1.$$

- (a) Derive the Yule–Walker equations $\gamma_k = \phi_1 \gamma_{k-1} + \phi_2 \gamma_{k-2}$ ($k \geq 1$) and the variance identity $\gamma_0 = \phi_1 \gamma_1 + \phi_2 \gamma_2 + \sigma^2$, making clear where *causality* is used. (2 pts)
- (b) Set up the 2×2 Yule–Walker system, invert it *by hand*, and report the estimates $\hat{\phi}_1, \hat{\phi}_2$ and $\hat{\sigma}^2$. (2 pts)
- (c) Verify that the fitted model is causal (stationary), e.g. by locating the roots of $\hat{\Phi}(z)$ or via the AR(2) stationarity triangle. (1 pt)

